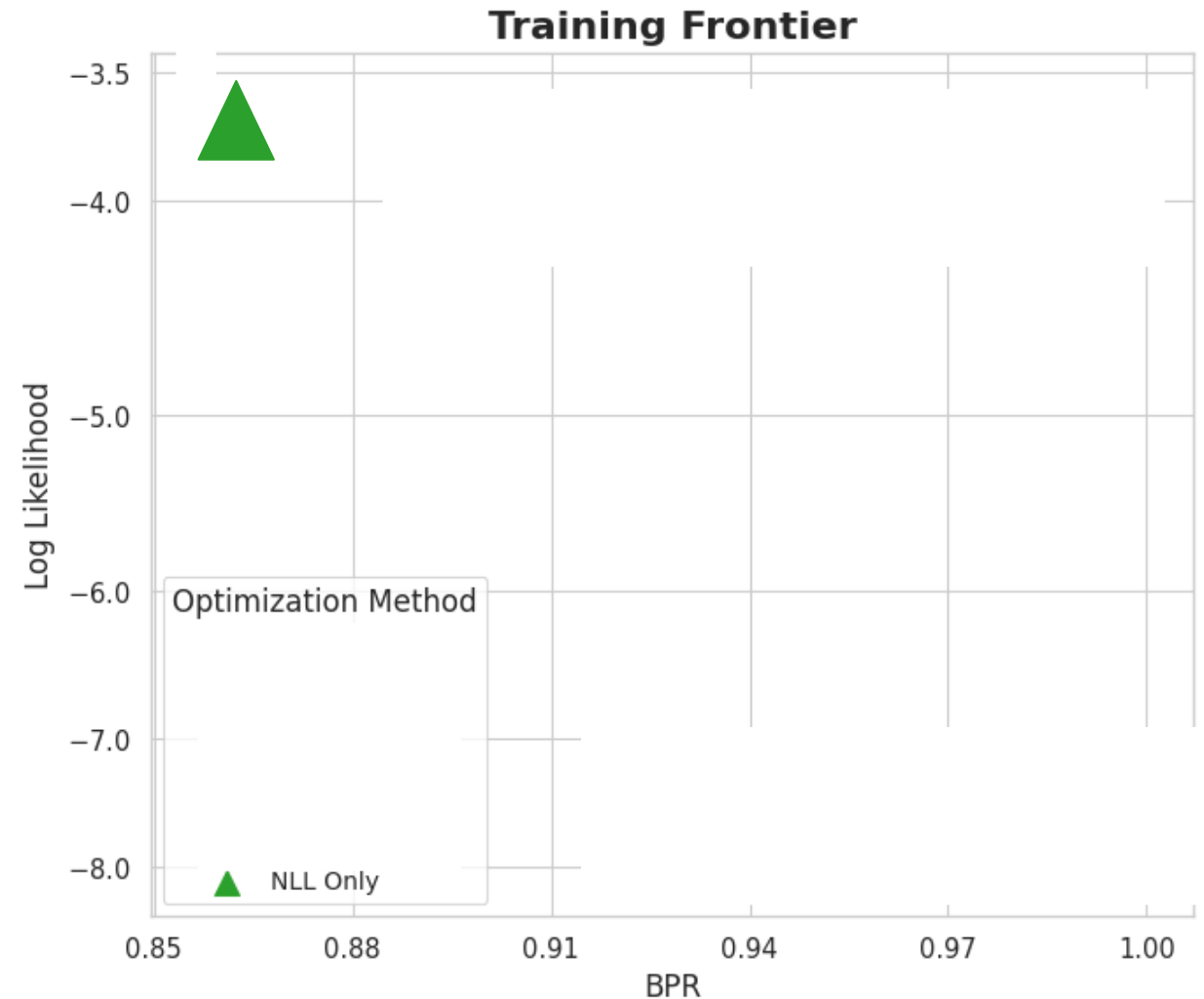


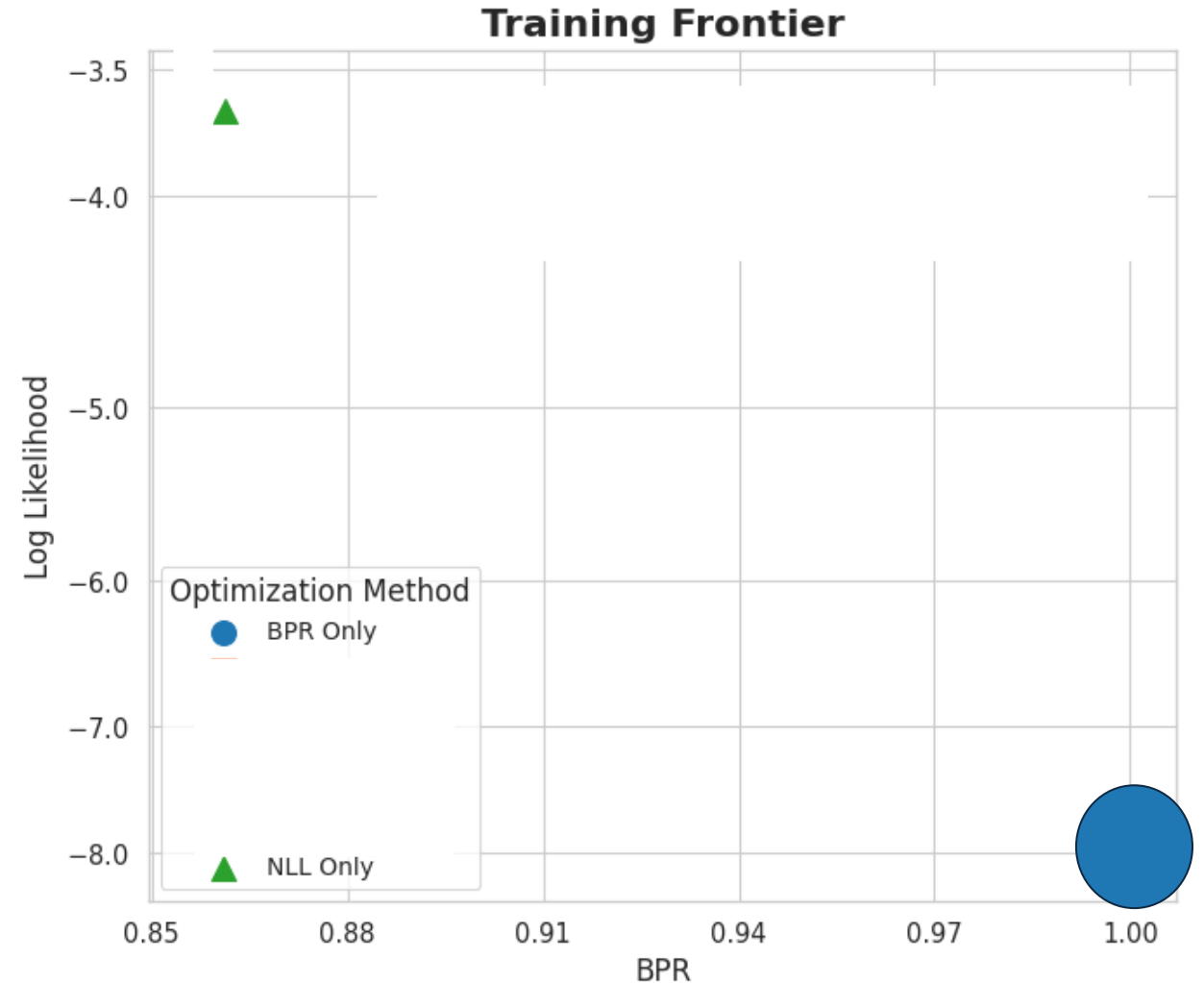
Contribution 3: How to train to balance BPR and Likelihood

$$\arg \max_{\phi} \sum_t \log p_{\phi}(y_t)$$



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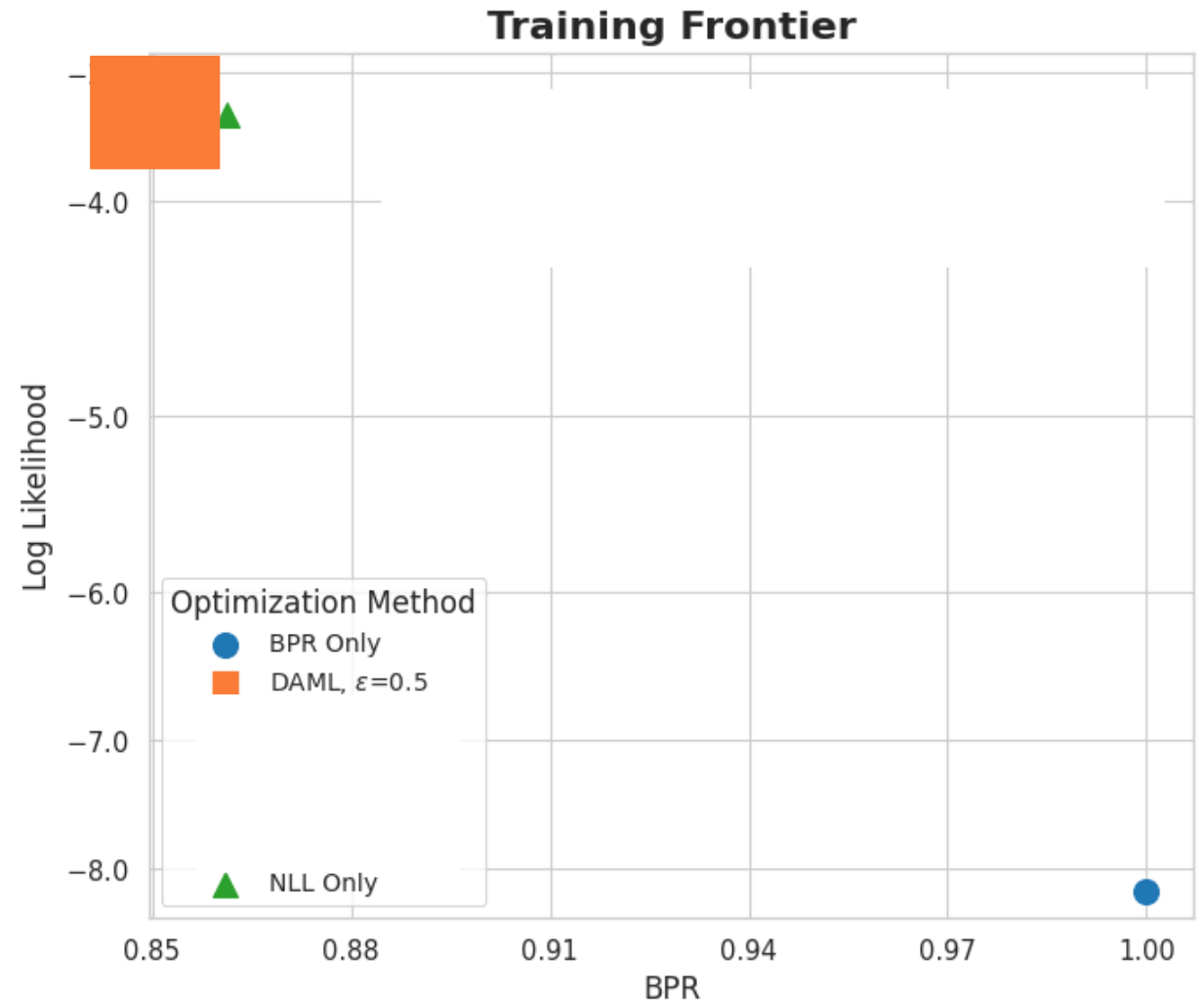
$$\arg \max_{\phi} \sum_t \text{BPR}(r_t(\phi), y_t)$$



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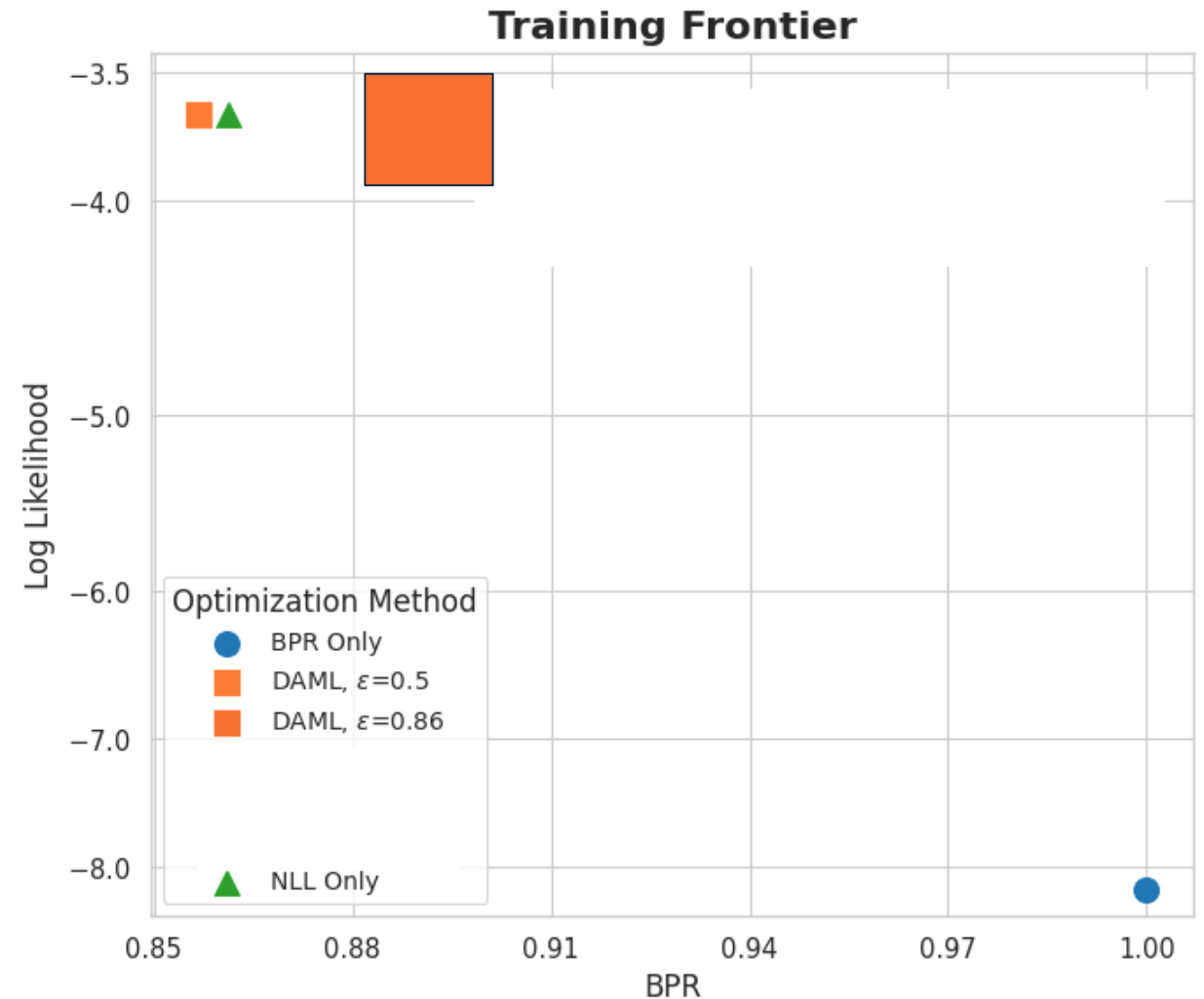
subj. to: $\text{BPR}(r_t(\phi), y_t) \geq \epsilon$ for all t



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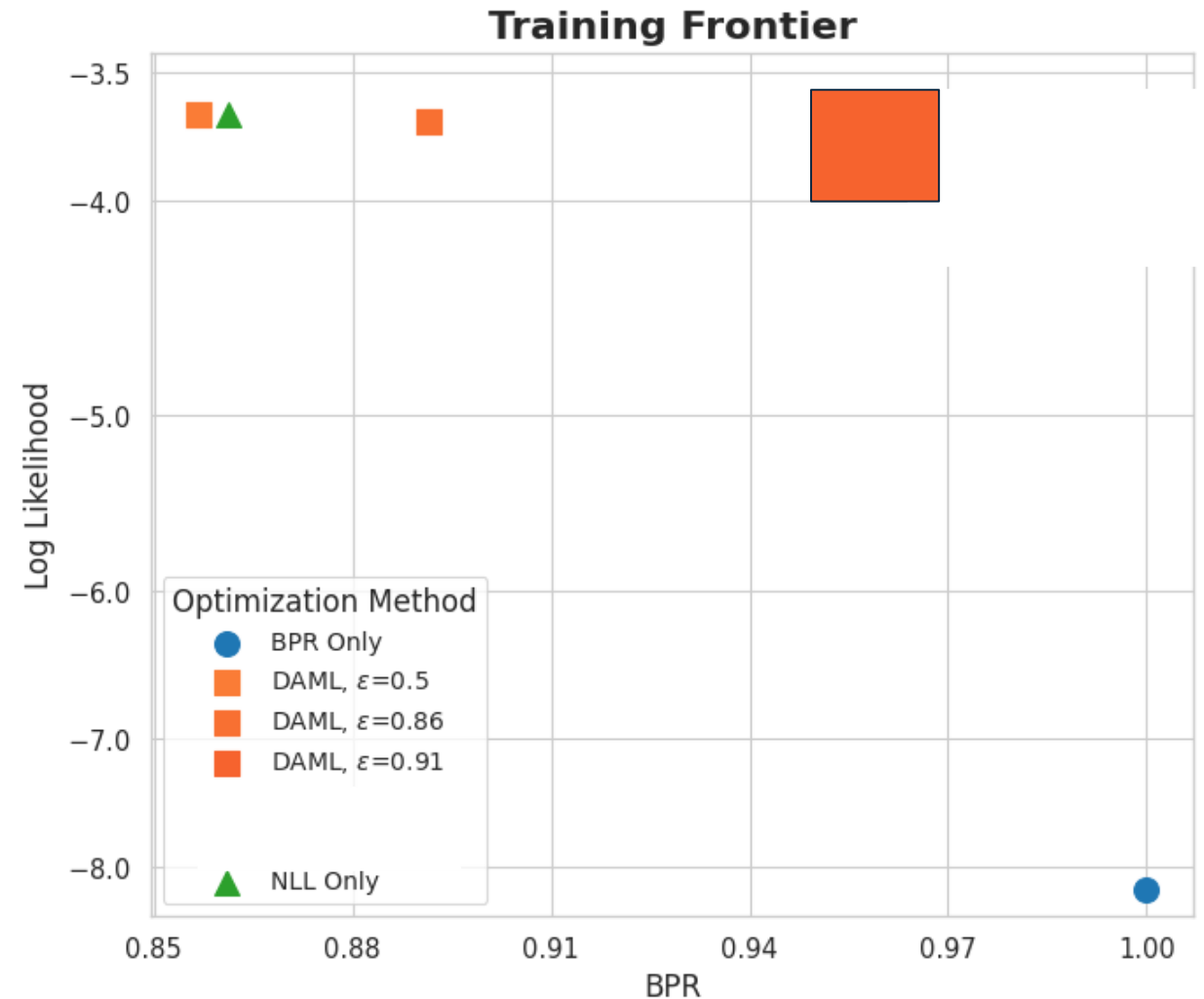
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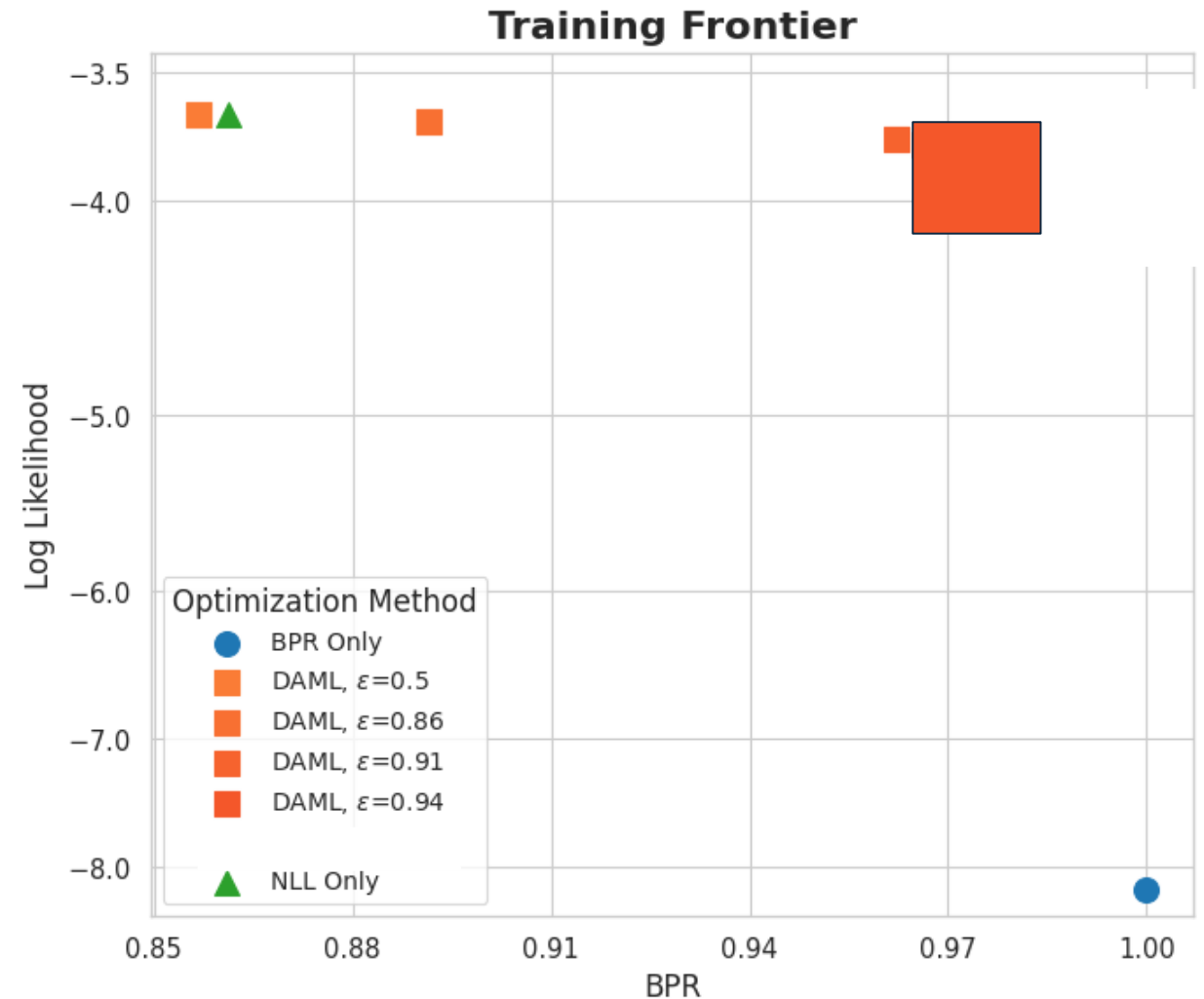
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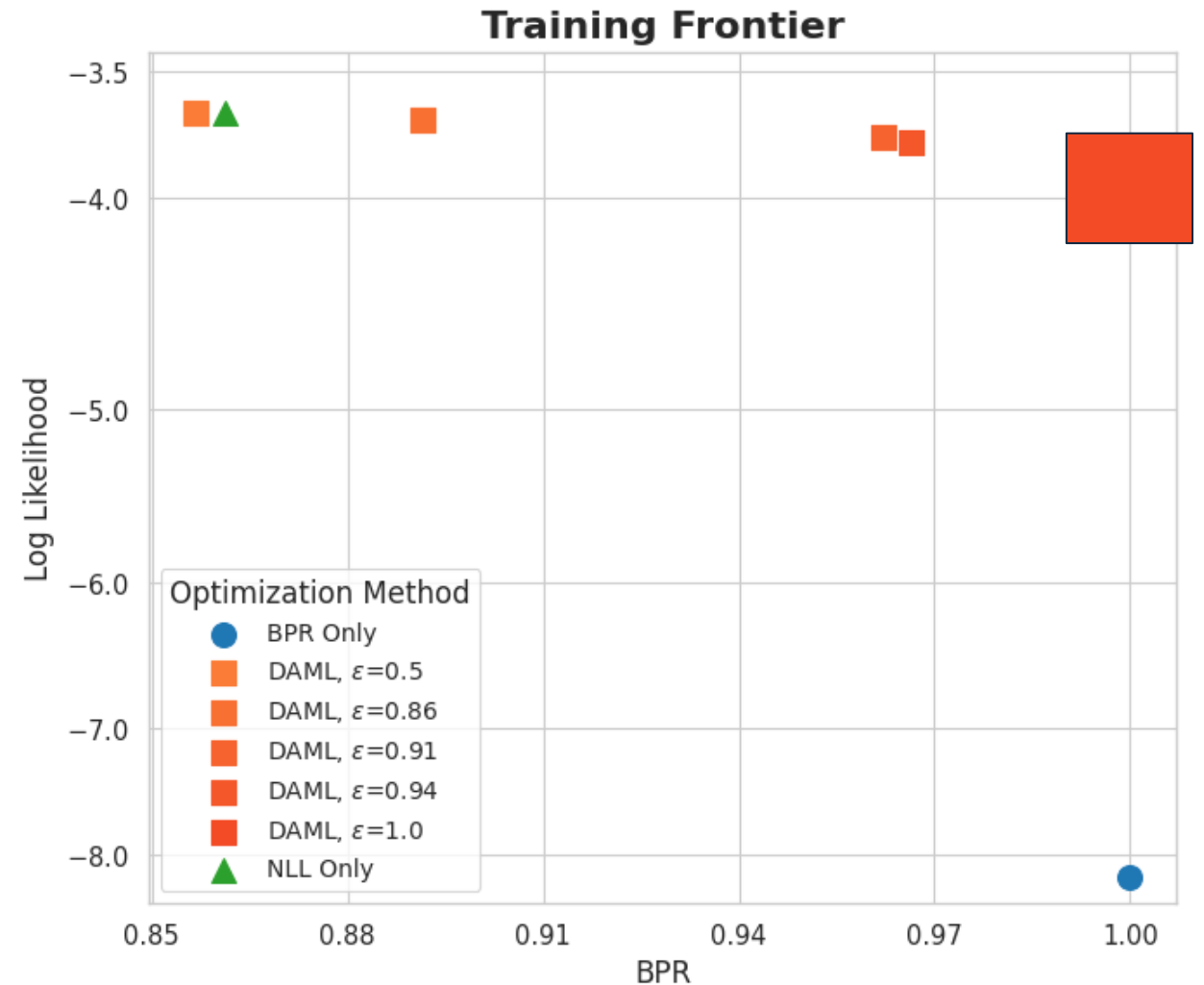
subj. to: $\text{BPR}(r_t(\phi), y_t) \geq \epsilon$ for all t



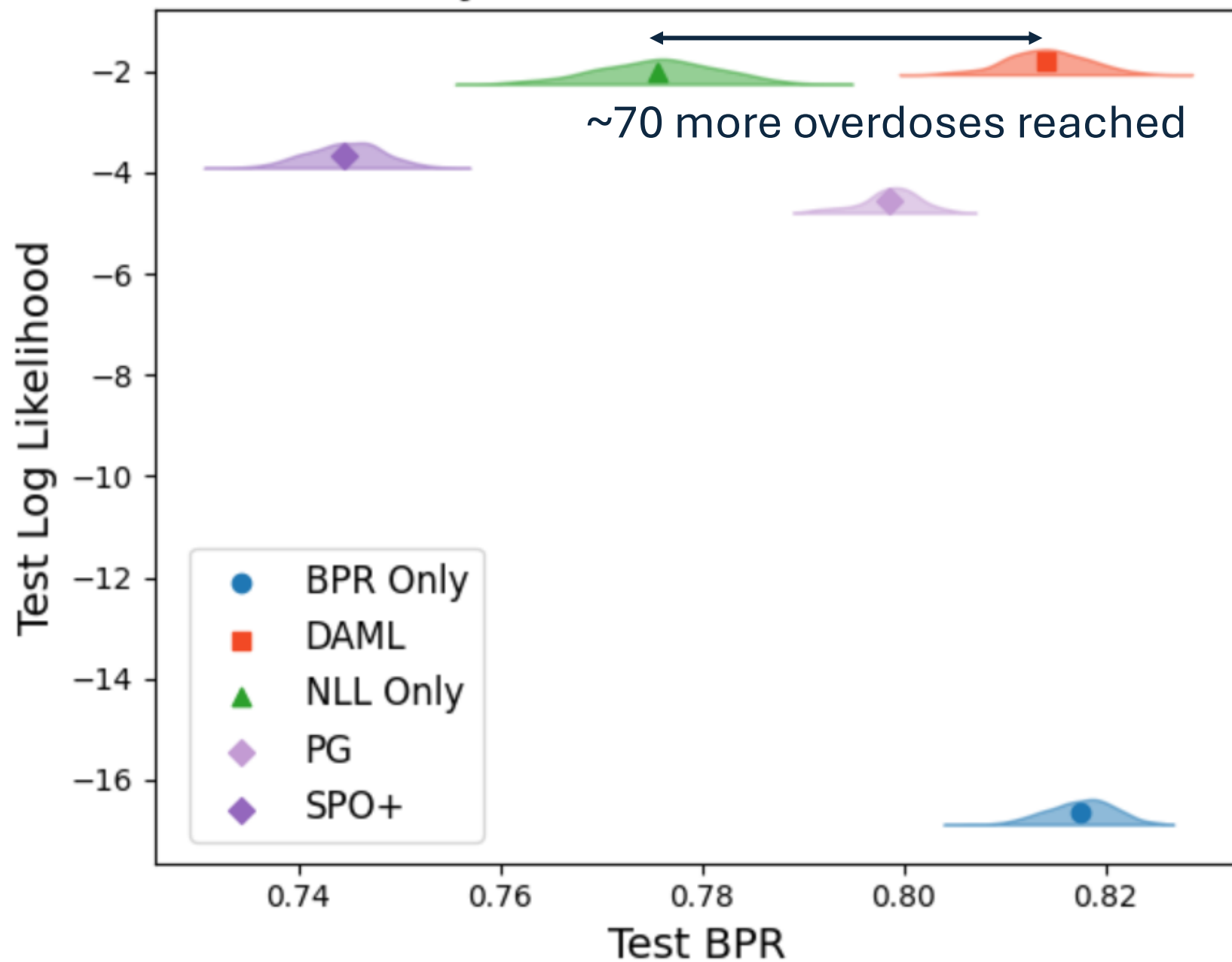
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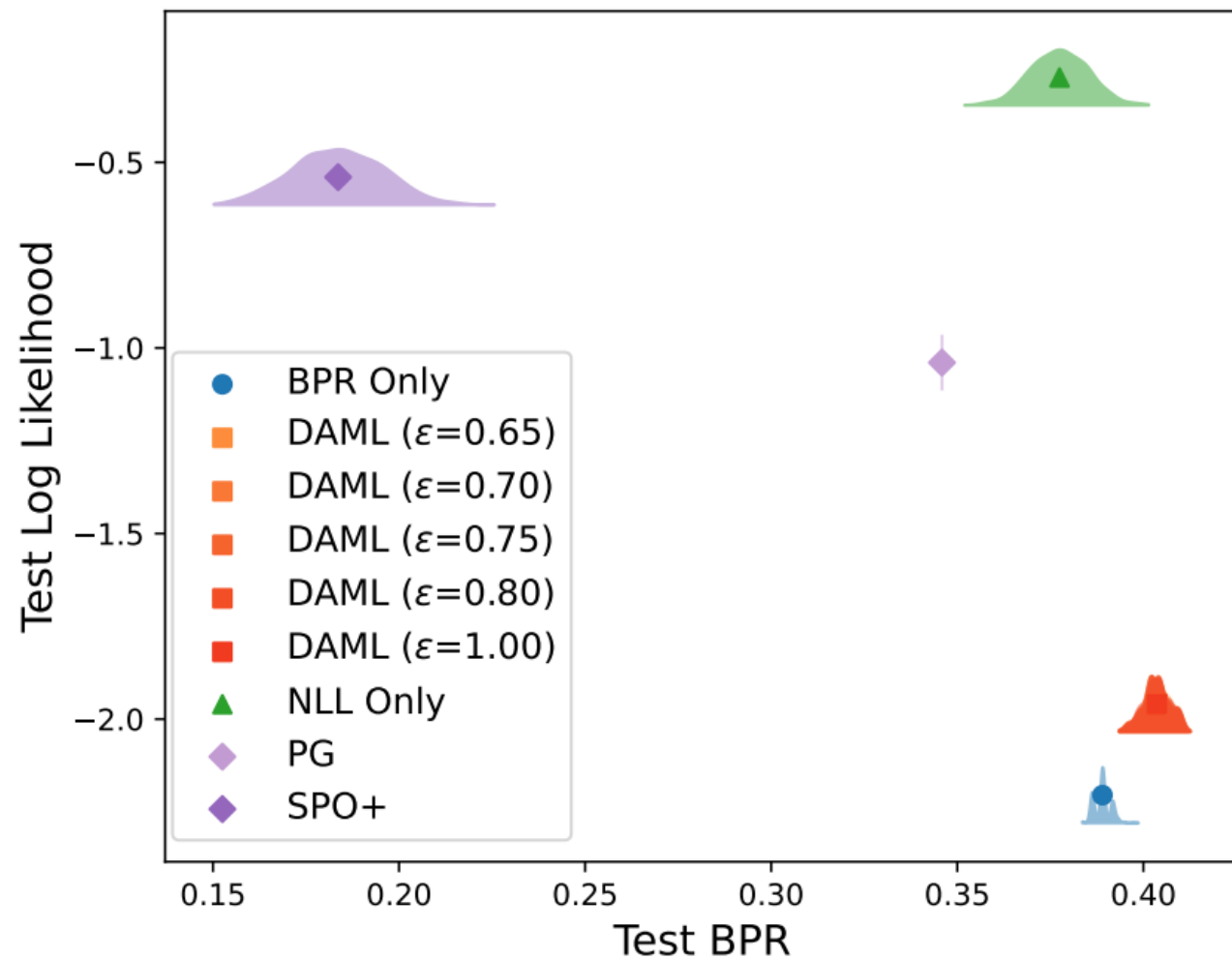
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Cook County IL Fatal Overdoses 2021-2022



ANWR TX Cranes, 2009-2010



MA Fatal Overdoses 2020-2021

